The Æther-flow Interpretation of Relativity

Higher-Derivative Quartic Compensator Branch Full Flat-Background Consistency and Reduction Test

Abstract

The previous branch-selection note chose the least-drift post-verdict candidate beyond the surviving quartic side program: adjoin one gapped auxiliary scalar Y so that the quartic block $(\Delta_\perp S)^2$ can be tuned into an exact square. That selection note proved only the isolated scalar subsection

$$\Delta\mathcal{L}^{(2)}_{\sigma y} = -\frac{m_S^2}{2}\big(\partial_0\sigma - \sigma_0\phi\big)^2 - \frac{1}{2}\left(\frac{\lambda_4}{M_*^2} - \frac{\alpha_Y^2}{M_Y^2}\right)(\nabla^2\sigma)^2,$$

and therefore left open the real post-selection burden: does the tuned branch still remove the observer correction after the existing Q-sector and longitudinal ordered-flow auxiliaries are reduced at the same scope?

This note performs that full flat-background quadratic reduction. The key new point is that the explicit nonlinear quartic completion uses

$$\Delta_\perp S \equiv \nabla_A\big(P^{AB}\nabla_B S\big)\,,$$

so at linear order one has not only $\nabla^2\sigma$ but

$$(\Delta_\perp S)^{(1)} = \nabla^2\sigma + \sigma_0\nabla^2(B+\chi),$$

where $h_{0i} = S_i + \partial_i B$ and $V_i = V_i^T + \partial_i\chi$. Thus the compensator branch mixes nontrivially with the longitudinal ordered-flow auxiliary χ, while the Q-sector remains exactly decoupled on the collapsed-scalar baseline.

After eliminating q^I, V_T^i, y, and χ, the full scalar bridge channel is

$$\mathcal{L}^{(2)}_{\hat\sigma,\mathrm{eff}} = -\frac{m_S^2}{2}\big(\partial_0\hat\sigma - \sigma_0\Phi_h\big)^2 - \frac{\Lambda_{\mathrm{red}}}{2}(\nabla^2\hat\sigma)^2, \qquad \Lambda_{\mathrm{red}} = \frac{\kappa_\theta\Lambda_Y}{\kappa_\theta + \sigma_0^2\Lambda_Y},$$

with

$$\Lambda_Y \equiv \frac{\lambda_4}{M_*^2} - \frac{\alpha_Y^2}{M_Y^2}, \qquad \Phi_h \equiv \phi + \partial_0\nabla^{-2}\left(\nabla^2 B - \frac{1}{2}\partial_0 H\right).$$

Hence the full reduced observer correction is

$$\Delta\mathcal{L}^{(2)}_{\mathrm{metric},4Y} = -\frac{1}{2}\Pi_\Phi^{(4Y)}(\omega,\mathbf{k})\,|\Phi_h|^2, \qquad \Pi_\Phi^{(4Y)} = m_S^2\sigma_0^2\frac{\Lambda_{\mathrm{red}}k^4}{m_S^2\omega^2 + \Lambda_{\mathrm{red}}k^4}.$$

The isolated subsection was therefore incomplete: the untuned quartic coefficient is renormalized by the longitudinal solve. But the tuned exact-square branch still passes the full test, because

$$\frac{\alpha_Y^2}{M_Y^2} = \frac{\lambda_4}{M_*^2} \quad\Longrightarrow\quad \Lambda_Y = \Lambda_{\mathrm{red}} = 0,$$

so the full reduced observer correction vanishes identically, not only its isolated ϕ-subsection. The Q-sector remains $q^I = 0$ on the decaying branch, Y stays algebraic and uniformly gapped by $M_Y^2 > 0$, and no new unsuppressed low-energy nonmetric mode is created by the compensator itself. Single-metric matter coupling is unchanged on the full branch.

The price of this success is structural and explicit. On the tuned branch the quartic observer scalar channel disappears completely, so the first replacement local/coercive architecture is no longer the old Einstein–quartic-scalar one. The narrowest replacement is instead a tuned-compensator strict auxiliary-elliptic branch: Q^I remain uniformly gapped slice-wise elliptic variables, Y is algebraic, (χ, W_i^T) are solved slice by slice under $\kappa_\theta > 0$, $\kappa_\Omega > 0$, and the scalar sector is carried only by the Stueckelberg-like momentum variable rather than by a quartic coercive Schrödinger channel.

1. Introduction

The quartic exact-square compensator branch was selected for one precise reason. The surviving quartic branch had already passed the matter-coupling gate, the recovery-compatibility gate, the admissible-background route, and the same-class nonlinear-stability route, yet its reduced observer equation still carried the nonzero scalar self-energy $\Pi_\phi^{(4)}$. If exact derivation remains the active goal, the next branch must therefore do more than suppress that remainder. It must remove it.

The previous compensator note chose the smallest admissible redesign and proved a first positive scalar-subsection result. But the project state recorded there also fixed the next burden explicitly: carry the compensator branch through the *full* flat-background quadratic reduction, including the existing Q-sector and ordered-flow auxiliaries, before treating the branch as genuinely upgraded. That is the sole purpose of the present note.

Framework and claim boundary. Æther / Æther-flow framework denotes the broader Æther / Æther-flow framework built on the claims that the Æther is the underlying four-dimensional substrate of reality, the Æther-flow is its intrinsic ordered motion, observed three-dimensional space is the local experiential slice of that deeper substrate, S-time is the experienced order of change arising from matter, light, and the Æther-flow, and observed expansion is the three-dimensional appearance of deeper four-dimensional ordered motion. Within that framework, Æther-flow Interpretation of Relativity denotes the exact-closure theory in which the effective gravitational dynamics are adopted to be exactly Einsteinian with universal matter coupling. In the active sequence, *adoption* means use of that established relativistic dynamics without claiming substrate derivation, while *derivation* is reserved for a first-principles recovery from explicit substrate variables.

The present note stays within that claim boundary. It does not claim a completed nonlinear theorem for the compensator branch. It does not claim a completed substrate derivation beyond flat-background quadratic scope. It does perform the exact next coefficient-level task forced by the selection note: full auxiliary elimination, full reduced observer correction, and a precise verdict on whether the tuned branch really removes the recorded observer remainder rather than only an isolated scalar slice of it.

2. Branch Data and the Missing Linearized Mixing

Definition 1 (Compensator branch data)**.** *The flat-background compensator branch keeps the bridge metric and matter route*

$$\mathfrak{g}_{AB} = G_{AB} - 2U_A U_B, \qquad S_{\text{matter}}[\Xi^*\mathfrak{g}, \psi] = S_{\text{matter}}[g, \psi], \tag{1}$$

keeps the collapsed-scalar baseline

$$\kappa_S = \mu_{SI} = \mu_S^2 = 0, \qquad m_S^2 > 0, \qquad \widehat{M}_{IJ} > 0, \tag{2}$$

and replaces the quartic scalar block by

$$\mathcal{L}_{4Y} = -\frac{\lambda_4}{2M_*^2}(\Delta_\perp S)^2 + \alpha_Y\, Y\, \Delta_\perp S - \frac{M_Y^2}{2}Y^2, \qquad M_Y^2 > 0, \tag{3}$$

with

$$\Delta_\perp S \equiv \nabla_A\Big(P^{AB}\nabla_B S\Big), \qquad P^{AB} = G^{AB} - U^A U^B. \tag{4}$$

Use the same admissible homogeneous flat background as in the previous linearized notes:

$$\bar{G}_{AB} = \delta_{AB}, \qquad \bar{U}^A = \delta^A{}_0, \qquad \bar{S} = \sigma_0 X^0, \qquad \bar{Q}^I = 0, \qquad \bar{Y} = 0. \tag{5}$$

Write perturbations as

$$G_{AB} = \delta_{AB} + H_{AB}, \qquad U^A = \bar{U}^A + V^A, \qquad S = \sigma_0 X^0 + \sigma, \qquad Q^I = q^I, \qquad Y = y, \tag{6}$$

with

$$V^0 = -\frac{1}{2}H_{00}, \qquad h_{00} = -2\phi = -H_{00}, \qquad h_{0i} = S_i + \partial_i B = -H_{0i} - 2V_i, \qquad h_{ij} = H_{ij}, \tag{7}$$

and

$$V_i = V_i^T + \partial_i\chi, \qquad \partial^i V_i^T = 0, \qquad H \equiv \delta^{ij}h_{ij}. \tag{8}$$

Proposition 1 (Full linearized projected divergence)**.** *On the flat background* (5)*, the linearized compensator source is*

$$(\Delta_\perp S)^{(1)} = \nabla^2\sigma + \sigma_0\nabla^2(B + \chi). \tag{9}$$

In particular, the compensator branch mixes at quadratic order with the longitudinal ordered-flow auxiliary χ, while no new linear mixing with the Q-sector appears on the collapsed-scalar baseline.

Proof. Define $X^A \equiv P^{AB}\nabla_B S$. On the background (5), one has $\bar{X}^A = 0$ because $\bar{P}^{A0} = 0$ and $\partial_i\bar{S} = 0$. Therefore

$$(\Delta_\perp S)^{(1)} = \partial_A(\delta X^A).$$

Since $\delta X^A = \bar{P}^{AB}\partial_B\sigma + \sigma_0\,\delta P^{A0}$, it remains to compute δP^{A0}. The unit constraint gives $V^0 = -H_{00}/2$, hence $\delta P^{00} = 0$. For spatial indices,

$$\delta P^{i0} = -H_{0i} - V_i = h_{0i} + V_i = S_i + V_i^T + \partial_i(B + \chi).$$

Taking the divergence and using $\partial^i S_i = \partial^i V_i^T = 0$ gives

$$\partial_A \delta P^{A0} = \nabla^2 (B + \chi).$$

Together with $\bar{P}^{ij}\partial_i\partial_j\sigma = \nabla^2\sigma$, this proves (9). The collapsed-scalar baseline (2) sets $\mu_{SI} = 0$, and (3) contains no q^I, so there is no new linear S/Q mixing. □

Remark 1. *This is the missing piece in the branch-selection note. The isolated subsection $(\Delta_\perp S)^{(1)} = \nabla^2\sigma$ is correct only after suppressing the scalar shift/longitudinal combination $B + \chi$. The full flat reduction must keep* (9).

3. Complete Quadratic Auxiliary Action

Proposition 2 (Full flat-background quadratic auxiliary Lagrangian)**.** *At fixed bridge perturbation $h_{\mu\nu}$, the complete quadratic nonmetric action of the compensator branch is*

$$\begin{aligned}\mathcal{L}^{(2)}_{\text{aux},4Y} = &-\frac{m_S^2}{2}(\partial_0\sigma - \sigma_0\phi)^2 - \frac{1}{2}q^I\Big((-\nabla^2)\delta_{IJ} + \widehat{M}_{IJ}\Big)q^J \\ &-\frac{\kappa_\theta}{2}\left(\nabla^2\chi + \frac{1}{2}\partial_0 H\right)^2 - \frac{\kappa_\Omega}{4}\,\partial_{[i}\Big(S_{j]} + V^T_{j]}\Big)\,\partial^{[i}\Big(S^{j]} + V_T^{j]}\Big) \\ &-\frac{\lambda_4}{2M_*^2}\,\Xi^2 + \alpha_Y\, y\,\Xi - \frac{M_Y^2}{2}y^2,\end{aligned} \tag{10}$$

where

$$\Xi \equiv \nabla^2\sigma + \sigma_0\nabla^2(B + \chi). \tag{11}$$

Proof. The kinetic, Q-sector, longitudinal ordered-flow, and transverse ordered-flow terms are exactly those already recorded on the collapsed quartic branch. Proposition 1 supplies the only new ingredient. Substituting (9) into (3) gives the last line of (10). □

Remark 2. *Proposition 2 answers the first question posed by the branch-selection note. The Q-sector remains decoupled, the transverse ordered-flow block is unchanged, and the only genuinely new mixing is the scalar-longitudinal pair (y, χ) through Ξ.*

4. Complete Auxiliary Elimination

4.1. Unchanged Q and Transverse Ordered-Flow Sectors

Proposition 3 (Exact Q-sector and transverse elimination)**.** *On the decaying flat branch with no harmonic zero mode and positive-definite $\widehat{M}_{IJ}$, the Q-sector and transverse ordered-flow equations remain*

$$q^I = 0, \qquad V_T^i = -S^i, \tag{12}$$

and contribute no residual quadratic observer correction.

Proof. The compensator term (3) contains neither q^I nor V_T^i. Therefore the Euler–Lagrange equations are unchanged from the earlier flat-background consistency notes. Positive-definite $\widehat{M}_{IJ}$ gives $q^I = 0$, and the vorticity block gives $V_T^i = -S^i$ on the decaying branch. □

4.2. Compensator Elimination

Definition 2 (Untuned compensator remainder)**.** *Define*

$$\Lambda_Y \equiv \frac{\lambda_4}{M_*^2} - \frac{\alpha_Y^2}{M_Y^2}. \tag{13}$$

The tuned exact-square branch is $\Lambda_Y = 0$.

Proposition 4 (Pointwise compensator solve)**.** *The compensator* y *is algebraic, with equation*

$$M_Y^2 y = \alpha_Y \, \Xi. \tag{14}$$

Substituting (14) *into* (10) *yields*

$$\begin{aligned}\mathcal{L}^{(2)}_{\sigma\chi,\mathrm{pre}} = & -\frac{m_S^2}{2}\big(\partial_0\sigma - \sigma_0\phi\big)^2 - \frac{\kappa_\theta}{2}\left(\nabla^2\chi + \frac{1}{2}\partial_0 H\right)^2 \\ & - \frac{\Lambda_Y}{2}\Big(\nabla^2\sigma + \sigma_0\nabla^2(B+\chi)\Big)^2 .\end{aligned} \tag{15}$$

Proof. Varying (10) with respect to y gives (14). Substitution gives

$$\alpha_Y y\Xi - \frac{M_Y^2}{2}y^2 = \frac{\alpha_Y^2}{2M_Y^2}\Xi^2,$$

which combines with the old quartic term into (15). □

Remark 3. *The compensator is therefore auxiliary already before the tuned limit. Its inverse quadratic operator is simply* M_Y^{-2}*, so the new field is uniformly gapped as long as* $M_Y^2 > 0$.

4.3. Longitudinal Ordered-Flow Elimination

Definition 3 (Metric scalar source for the longitudinal solve)**.** *On the decaying branch define*

$$\Psi_h \equiv \nabla^2 B - \frac{1}{2}\partial_0 H. \tag{16}$$

Proposition 5 (Full longitudinal elimination)**.** *Assume*

$$\kappa_\theta + \sigma_0^2\Lambda_Y \neq 0. \tag{17}$$

Then the longitudinal auxiliary χ *is determined by*

$$\nabla^2\chi = -\frac{\frac{\kappa_\theta}{2}\partial_0 H + \Lambda_Y\sigma_0\big(\nabla^2\sigma + \sigma_0\nabla^2 B\big)}{\kappa_\theta + \sigma_0^2\Lambda_Y}, \tag{18}$$

and substitution into (15) *yields*

$$\mathcal{L}^{(2)}_{\sigma,\mathrm{full}} = -\frac{m_S^2}{2}\big(\partial_0\sigma - \sigma_0\phi\big)^2 - \frac{\Lambda_{\mathrm{red}}}{2}\big(\nabla^2\sigma + \sigma_0\Psi_h\big)^2, \tag{19}$$

where

$$\Lambda_{\rm red} \equiv \frac{\kappa_\theta \Lambda_Y}{\kappa_\theta + \sigma_0^2 \Lambda_Y}. \tag{20}$$

Proof. Let

$$X \equiv \nabla^2 \chi, \qquad h \equiv \frac{1}{2}\partial_0 H, \qquad d \equiv \nabla^2 \sigma + \sigma_0 \nabla^2 B.$$

Then (15) becomes

$$\mathcal{L}^{(2)}_{\sigma\chi,\rm pre} = -\frac{m_S^2}{2}\left(\partial_0\sigma - \sigma_0\phi\right)^2 - \frac{1}{2}\left[\kappa_\theta (X+h)^2 + \Lambda_Y (d + \sigma_0 X)^2\right].$$

Variation with respect to χ gives

$$\nabla^2[\kappa_\theta(X+h) + \Lambda_Y \sigma_0 (d + \sigma_0 X)] = 0.$$

On the decaying branch with no harmonic zero mode the bracket vanishes, which is exactly (18). Substituting the resulting minimizing value of X gives

$$\kappa_\theta (X+h)^2 + \Lambda_Y (d+\sigma_0 X)^2 = \frac{\kappa_\theta \Lambda_Y}{\kappa_\theta + \sigma_0^2 \Lambda_Y}(d - \sigma_0 h)^2.$$

Since $d - \sigma_0 h = \nabla^2\sigma + \sigma_0 \Psi_h$, this proves (19)–(20). □

Remark 4. *The isolated scalar subsection of the branch-selection note saw Λ_Y; the full reduction sees $\Lambda_{\rm red}$. Longitudinal auxiliary elimination therefore genuinely renormalizes the untuned quartic coefficient. This is exactly the effect that had to be checked beyond the isolated subsection.*

Corollary 1 (Tuned branch and the longitudinal denominator)**.** *On the tuned exact-square branch $\Lambda_Y = 0$,*

$$\Lambda_{\rm red} = 0, \qquad \nabla^2 \chi = -\frac{1}{2}\partial_0 H. \tag{21}$$

In particular, the tuned branch does not change the longitudinal elliptic coefficient: it is exactly the old κ_θ solve.

Proof. Insert $\Lambda_Y = 0$ into (18) and (20). □

5. Full Reduced Observer Correction

Definition 4 (Full scalar metric combination)**.** *On the decaying branch with no harmonic zero mode, define*

$$\Phi_h \equiv \phi + \partial_0 \nabla^{-2}\Psi_h = \phi + \partial_0 \nabla^{-2}\left(\nabla^2 B - \frac{1}{2}\partial_0 H\right). \tag{22}$$

Proposition 6 (Field redefinition to the reduced scalar channel)**.** *Define*

$$\hat\sigma \equiv \sigma + \sigma_0 \nabla^{-2}\Psi_h. \tag{23}$$

Then (19) *becomes*

$$\mathcal{L}^{(2)}_{\hat\sigma,\rm eff} = -\frac{m_S^2}{2}\left(\partial_0\hat\sigma - \sigma_0\Phi_h\right)^2 - \frac{\Lambda_{\rm red}}{2}(\nabla^2\hat\sigma)^2. \tag{24}$$

Proof. By construction,

$$\nabla^2\hat{\sigma} = \nabla^2\sigma + \sigma_0\Psi_h,$$

which matches the second term of (19). Likewise

$$\partial_0\hat{\sigma} - \sigma_0\Phi_h = \partial_0\sigma + \sigma_0\partial_0\nabla^{-2}\Psi_h - \sigma_0\phi - \sigma_0\partial_0\nabla^{-2}\Psi_h = \partial_0\sigma - \sigma_0\phi.$$

Substituting these identities into (19) gives (24). □

Theorem 1 (Full reduced observer correction)**.** *After eliminating q^I, V_T^i, y, χ, and $\hat{\sigma}$, the full flat-background reduced observer correction in the scalar bridge sector is*

$$\Delta\mathcal{L}^{(2)}_{\mathrm{metric},4Y}(\omega,\mathbf{k}) = -\frac{1}{2}\Pi_\Phi^{(4Y)}(\omega,\mathbf{k})\,|\Phi_h(\omega,\mathbf{k})|^2, \tag{25}$$

with

$$\Pi_\Phi^{(4Y)}(\omega,\mathbf{k}) = m_S^2\sigma_0^2\frac{\Lambda_{\mathrm{red}}k^4}{m_S^2\omega^2 + \Lambda_{\mathrm{red}}k^4}. \tag{26}$$

Proof. Equation (24) is exactly the same Gaussian scalar structure already used in the earlier flat-background consistency notes, with ϕ replaced by Φ_h and the quartic coefficient replaced by Λ_{red}. Integrating out $\hat{\sigma}$ therefore yields the same tree-level self-energy formula, which is (25)–(26). □

Corollary 2 (Full tuned-branch removal of the observer correction)**.** *On the tuned exact-square branch*

$$\frac{\alpha_Y^2}{M_Y^2} = \frac{\lambda_4}{M_*^2}, \tag{27}$$

the full flat-background reduced observer correction vanishes identically:

$$\Pi_\Phi^{(4Y)}(\omega,\mathbf{k}) \equiv 0. \tag{28}$$

In particular, the tuned branch removes the recorded quartic observer remainder after complete *auxiliary elimination, not only on the isolated ϕ-subsection of the earlier selection note.*

Proof. Equation (27) is exactly $\Lambda_Y = 0$, so Corollary 1 gives $\Lambda_{\mathrm{red}} = 0$. Substituting this into (26) yields (28). □

Remark 5. *The full-reduction answer is sharper than the isolated subsection in two directions. Untuned branches are changed nontrivially by the longitudinal solve through Λ_{red}, so the isolated subsection was not yet the full answer. But the exact-square tuning survives that extra reduction cleanly because zero stays zero under the full solve.*

6. Mode Control and Matter Coupling on the Full Branch

Proposition 7 (No new unsuppressed compensator mode)**.** *Assume*

$$M_Y^2 > 0, \qquad \widehat{M}_{IJ} > 0, \qquad \kappa_\theta > 0, \qquad \kappa_\Omega > 0. \tag{29}$$

Then the compensator branch introduces no new unsuppressed low-energy nonmetric observer mode at the present flat-background quadratic scope:

1. *q^I remain gapped and auxiliary by Proposition 3;*

2. *y is algebraic with uniform gap M_Y^2 by Proposition 4;*

3. *the ordered-flow variables χ and V_T^i remain slice-wise auxiliary, with the only possible new singular locus given by $\kappa_\theta + \sigma_0^2 \Lambda_Y = 0$;*

4. *on the tuned branch $\Lambda_Y = 0$, that singular locus is absent and the χ-solve reduces exactly to the preexisting elliptic relation* (21).

Proof. Items (1) and (2) are Propositions 3 and 4. Item (3) is Proposition 5, which shows that the longitudinal elimination is controlled precisely by the denominator $\kappa_\theta + \sigma_0^2 \Lambda_Y$. Item (4) is Corollary 1. Therefore the compensator itself does not create a new propagating low-energy mode, and on the tuned branch the longitudinal auxiliary architecture is exactly the old one. □

Remark 6. *What survives on the tuned branch is not a new compensator mode. It is only the already-known collapsed-scalar Stueckelberg-like structure carried by $(\partial_0 \sigma - \sigma_0 \phi)^2$. The compensator removes the quartic observer correction but does not by itself convert that old degenerate scalar limit into a new healthy propagating nonmetric channel.*

Theorem 2 (Single-metric matter coupling on the full compensator branch)**.** *At the present action-level and flat-background quadratic scope, the compensator branch preserves controlled matter coupling through the bridge metric alone. On the tuned branch* (27)*, the reduced observer-accessible action is*

$$S^{(2)}_{\rm obs}[h, \psi] = S^{(2)}_{\rm EH}[h] + S^{(2)}_{\rm matter}[h, \psi], \tag{30}$$

with no residual nonmetric observer correction at this scope.

Proof. The action-level matter route is unchanged by Definition 1: matter depends only on $g_{\mu\nu} = \Xi^* \mathfrak{g}_{AB}$. At fixed $h_{\mu\nu}$, the fields $(q^I, \chi, V_T^i, \sigma, y)$ appear only in the auxiliary sector (10). Therefore there is no direct matter dependence on any nonmetric field beyond the metric itself. The complete auxiliary elimination performed above yields the reduced metric correction (25). On the tuned branch, Corollary 2 shows that this correction vanishes identically. Hence (30) follows. □

7. Narrowest Replacement Local/Coercive Architecture

The tuned branch passes the full flat-background reduction test, but it changes the dynamical architecture. The old quartic observer scalar channel is gone by design. The replacement local/coercive setup should therefore preserve only what survives that removal.

Definition 5 (Tuned-compensator strict auxiliary-elliptic branch)**.** *The* tuned-compensator strict auxiliary-elliptic branch *is the branch defined by*

$$\frac{\alpha_Y^2}{M_Y^2} = \frac{\lambda_4}{M_*^2}, \qquad M_Y^2 > 0, \qquad \widehat{M}_{IJ} > 0, \qquad \kappa_\theta > 0, \qquad \kappa_\Omega > 0, \tag{31}$$

in unit-lapse spatial-harmonic gauge. On this branch, Q^I remain the same uniformly positive elliptic/gapped sector as before, Y is solved algebraically, and (χ, W_i^T) are solved slice by slice by the same weighted/homogeneous elliptic package already used on the fixed-completion unit-lapse route.

Proposition 8 (Narrowest replacement architecture)**.** *At the present disciplined scope, the narrowest replacement local/coercive architecture for the tuned compensator branch is the one of Definition 5. Concretely:*

1. *the bridge metric, matter route, and unit-lapse weighted/homogeneous longitudinal-control package are kept unchanged;*
2. *the Q-sector is kept as the same uniformly gapped elliptic block and the compensator Y as an algebraic slice-wise solve rather than as a new propagating channel;*
3. *the ordered-flow auxiliary equations are unchanged at principal level from the repaired unit-lapse route because $\Lambda_Y = 0$ restores the old χ-solve exactly;*
4. *the scalar control variable is no longer the quartic Schrödinger energy $\|\Delta_\gamma \sigma\|^2$, which has been intentionally neutralized, but the Stueckelberg-like momentum variable $\pi_\sigma = m_S^2(n^\mu \nabla_\mu S - \sigma_0)$ together with the elliptically controlled auxiliary package.*

Proof. Items (1)–(3) follow directly from Corollary 1, Proposition 7, and the fact that the compensator branch does not alter the bridge metric or matter route. Item (4) is forced by Corollary 2: once the tuned branch removes the quartic operator entirely, the old quartic coercive scalar channel no longer exists to support the previous Einstein–quartic-scalar architecture. The smallest honest replacement is therefore to keep the repaired auxiliary-elliptic gauge package that still applies, while downgrading the scalar sector to the surviving Stueckelberg-like momentum structure rather than inventing a new propagating nonmetric principal term. □

Remark 7. *Proposition 8 is a programmatic reduction verdict, not yet a theorem of local well-posedness. Its purpose is narrower: to state the smallest replacement architecture now justified by the full quadratic reduction. The next substantive manuscript, if this branch remains the active derivational candidate, should therefore be an explicit tuned-compensator completion/reduced-system note rather than another asymptotic note on the old quartic side program.*

8. Conclusion

The immediate task after the compensator branch-selection note is now complete.

1. The full flat-background quadratic action of the compensator branch has been derived, and the missing longitudinal mixing is explicit through $(\Delta_\perp S)^{(1)} = \nabla^2\sigma + \sigma_0 \nabla^2(B + \chi)$.
2. The Q-sector remains exactly decoupled on the collapsed-scalar baseline, while the longitudinal ordered-flow auxiliary renormalizes the untuned quartic coefficient from Λ_Y to $\Lambda_{\rm red}$.
3. After complete auxiliary elimination, the full reduced observer correction is the scalar-metric self-energy $\Pi_\Phi^{(4Y)}$ of (26).
4. On the tuned exact-square branch, $\Lambda_Y = \Lambda_{\rm red} = 0$, so the full reduced observer correction vanishes identically. The tuned branch therefore really removes the recorded $\Pi_\phi^{(4)}$ at the full reduction level.
5. The compensator remains algebraic and uniformly gapped, the branch introduces no new unsuppressed low-energy compensator mode, and single-metric matter coupling is untouched on the full branch.

6. The narrowest replacement local/coercive setup is no longer the old Einstein–quartic-scalar architecture but the tuned-compensator strict auxiliary-elliptic branch, in which the repaired unit-lapse weighted/homogeneous longitudinal package is preserved while the scalar channel is reduced to the Stueckelberg-like momentum structure.

At the present disciplined scope, this is the correct post-selection verdict. The compensator branch survives the full quadratic reduction test and remains the active least-drift candidate beyond the old quartic side program. The next burden is therefore not another asymptotic repair of that old side program. It is to choose one explicit tuned-compensator nonlinear completion and write its first reduced system in the replacement auxiliary-elliptic architecture just identified here.